

A089 · Tracing oncogene amplification and genome architecture across single-cell tumor phylogenies

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Session: Day 1 · Thu Oct 1, 2026 · 2:15–4:15 PM

About one-third of adult solid tumors are defined by focal amplification of an essential oncogenic driver — ERBB2, MYCN, EGFR, FGFR2, CCNE1, or MYC across diverse lineages. These amplification-driven cancers are among the most therapeutically challenging adult malignancies, despite advances in targeted therapy and immunotherapy. Focal amplifications take structurally distinct forms: integrated amplicons (tandem duplications, complex rearrangements, chromothripsis) are inherited according to Mendelian inheritance while extra chromosomal DNA (ecDNA) are circular, centromere-free elements that segregate randomly, driving copy-number heterogeneity and rapid remodeling under selection. Furthermore, the structural architecture of an amplification strongly influences how amplifications evolve and respond to selection. However, whether subclonal trends in amplicon architecture impact cellular fitness or steer a tumor's evolutionary trajectory has not been systematically addressed.

Using published scWGS from ovarian cancer samples, we look to quantify amplification clonality, structural heterogeneity, and antigen persistence across tumor phylogenies. This work will focus on ovarian cancer patients with CCNE1 amplifications; these patients are part of a homologous-recombination proficient subtype that has no second-line treatment options after chemotherapy. Our analysis quantifies whether nominated surface antigens remain co-amplified and co-expressed with essential drivers throughout tumor evolution or instead diverge through loss of ecDNA, segmental deletion, or transcriptional silencing. Ultimately, this resolves amplicon topology to determine whether cancer genome architecture is an evolutionary mechanism for therapeutic escape.